Test framebreaks

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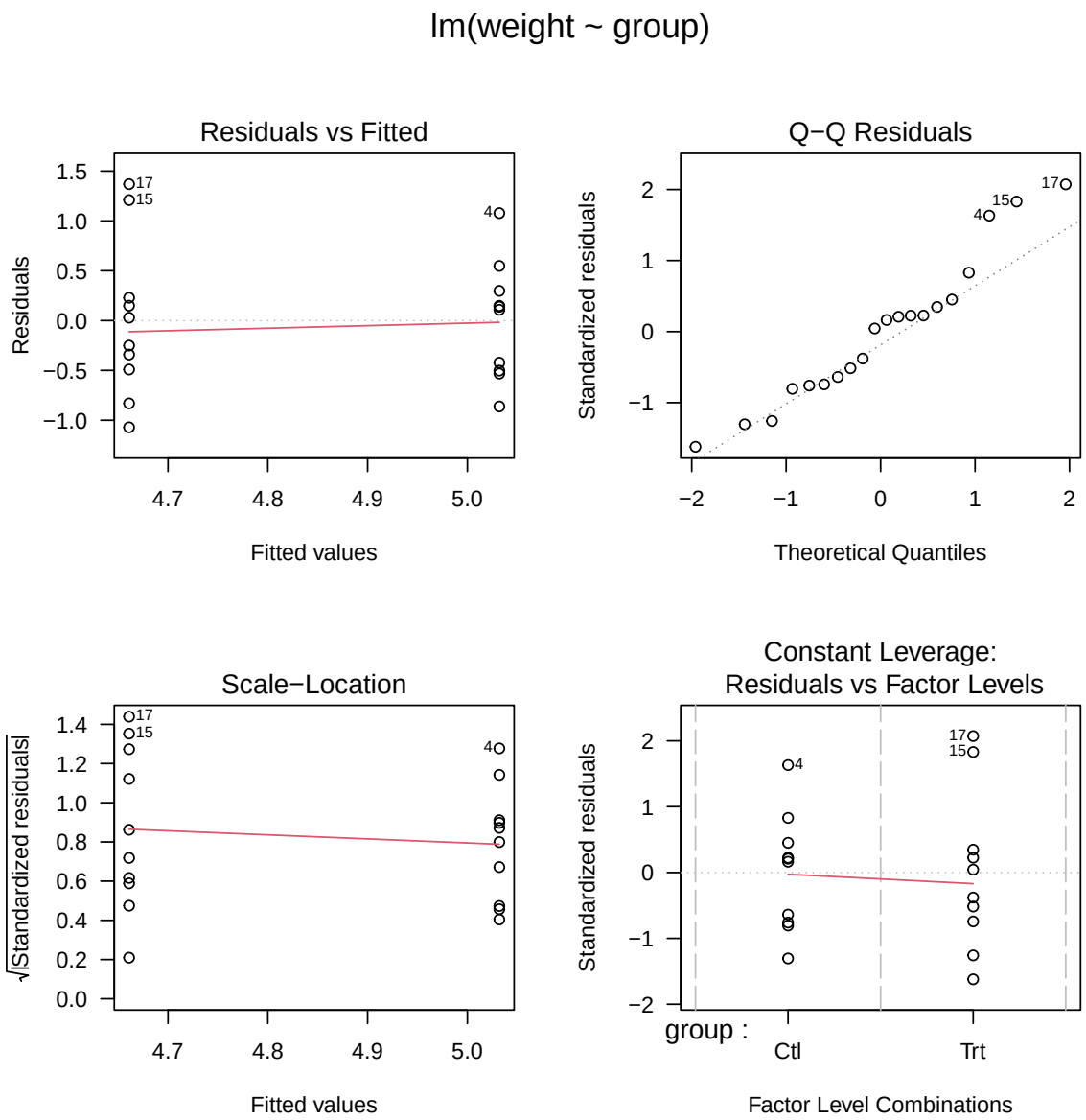
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Preliminaries I

```
example(step, run.donttest = TRUE)

##
## step> ## No test:
## step> ## following on from example(lm)
## step> ## Don't show:
## step> utils::example("lm", echo = FALSE)
```

Preliminaries II



Preliminaries III

```
##
## step> ## End(Don't show)
## step> step(lm.D9)
## Start:  AIC=-12.58
## weight ~ group
##
##         Df Sum of Sq  RSS   AIC
## - group  1     0.688 9.42 -13.1
## <none>               8.73 -12.6
##
## Step:  AIC=-13.06
## weight ~ 1
##
##
## Call:
## lm(formula = weight ~ 1)
##
## Coefficients:
## (Intercept)
##        4.85
```

Preliminaries IV

```
##
##
## step> summary(lm1 <- lm(Fertility ~ ., data = swiss))
##
## Call:
## lm(formula = Fertility ~ ., data = swiss)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -15.274  -5.262   0.503   4.120  15.321
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)       66.9152    10.7060    6.25  1.9e-07 ***
## Agriculture       -0.1721     0.0703   -2.45   0.0187 *
## Examination       -0.2580     0.2539   -1.02   0.3155
## Education         -0.8709     0.1830   -4.76  2.4e-05 ***
## Catholic           0.1041     0.0353    2.95   0.0052 **
## Infant.Mortality   1.0770     0.3817    2.82   0.0073 **
## ---
```

Preliminaries V

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.17 on 41 degrees of freedom
## Multiple R-squared:  0.707, Adjusted R-squared:  0.671
## F-statistic: 19.8 on 5 and 41 DF,  p-value: 5.59e-10
##
##
## step> slm1 <- step(lm1)
## Start:  AIC=190.7
## Fertility ~ Agriculture + Examination + Education + Catholic +
##     Infant.Mortality
##
##                    Df Sum of Sq  RSS AIC
## - Examination       1        53 2158 190
## <none>                          2105 191
## - Agriculture       1       308 2413 195
## - Infant.Mortality  1       409 2514 197
## - Catholic          1       448 2553 198
## - Education         1      1163 3268 209
##
```

Preliminaries VI

```
## Step:  AIC=189.9
## Fertility ~ Agriculture + Education + Catholic + Infant.Mortality
##
##                    Df Sum of Sq  RSS AIC
## <none>                          2158 190
## - Agriculture       1       264 2422 193
## - Infant.Mortality  1       410 2568 196
## - Catholic          1       957 3115 205
## - Education         1      2250 4408 221
##
## step> summary(slm1)
##
## Call:
## lm(formula = Fertility ~ Agriculture + Education + Catholic +
##     Infant.Mortality, data = swiss)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -14.676  -6.052   0.751   3.166  16.142
##
```

Preliminaries VII

```
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)       62.1013     9.6049    6.47  8.5e-08 ***
## Agriculture       -0.1546     0.0682   -2.27   0.0286 *
## Education         -0.9803     0.1481   -6.62  5.1e-08 ***
## Catholic           0.1247     0.0289    4.31  9.5e-05 ***
## Infant.Mortality   1.0784     0.3819    2.82   0.0072 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.17 on 42 degrees of freedom
## Multiple R-squared:  0.699, Adjusted R-squared:  0.671
## F-statistic: 24.4 on 4 and 42 DF,  p-value: 1.72e-10
##
##
## step> slm1$anova
##            Step Df Deviance Resid. Df Resid. Dev   AIC
## 1              NA       NA        41       2105 190.7
## 2 - Examination  1    53.03        42       2158 189.9
##
```

Preliminaries VIII

```
## step> ## End(No test)
## step>
## step>
```